could not. There is only a limited amount of change that can be achieved by direct transformation in the 'swords to ploughshares' manner. Really radical change can be achieved only by going 'back to the drawing board', throwing away the previous design and starting afresh. When engineers go back to the drawing board and create a new design, they do not necessarily throw away the ideas from the old design. But they don't literally try to deform the old physical object into the new one. The old object is too weighed down with the clutter of history. Maybe you can beat a sword into a ploughshare, but try 'beating' a propellor engine into a jet engine! You can't do it. You have to discard the propellor engine and go back to the drawing board.

Living things, of course, were never designed on drawing boards. But they do go back to fresh beginnings. They make a clean start in every generation. Every new organism begins as a single cell and grows anew. It inherits the *ideas* of ancestral design, in the form of the DNA program, but it does not inherit the physical organs of its ancestors. It does not inherit its parent's heart and *remould* it into a new (and possibly improved) heart. It starts from scratch, as a single cell, and grows a new heart, using the same design program as its parent's heart, to which improvements may be added. You see the conclusion I am leading up to. One important thing about a 'bottlenecked' life cycle is that it makes possible the equivalent of going back to the drawing board.

Bottlenecking of the life cycle has a second, related consequence. It provides a 'calendar' that can be used to regulate the processes of embryology. In a bottlenecked life cycle, every fresh generation marches through approximately the same parade of events. The organism begins as a single cell. It grows by cell division. And it reproduces by sending out daughter cells. Presumably it eventually dies, but that is less important than it seems to us mortals; as far as this discussion is concerned the end of the cycle is reached when the present organism reproduces and a new generation's cycle begins. Although in theory the organism could reproduce at any time during its growth phase, we can expect that eventually an optimum time for reproduction would emerge. Organisms that released spores when they were too young or too old would end up with fewer descendants than rivals that built up their strength and then released a massive number of spores when in the prime of life.